Multi-Agent Patterns, Workflows and Orchestration

Instructions: Choose the best answer for each question. The answer key is printed upside down at the bottom of the last page.

1. In a multi-agent workflow, what does a node usually represent?

A. A connection between two agents
B. A unit of computation, such as a function, agent, or workflow step
C. A termination condition
D. A user interface component

2. Which workflow pattern is best described as A -> B -> C, where each step must finish before the next one starts?

A. Parallel workflow
B. Conditional workflow
C. Sequential workflow
D. Handoff workflow

3. In the automated ML assignment review example, why is the workflow sequential?

A. Because all agents speak at the same time
B. Because each step depends on the output of the previous step
C. Because the system chooses the next step randomly
D. Because the workflow has no input data

4. What is the main purpose of a conditional workflow?

A. To execute all steps at the same time
B. To route execution based on logic or previous step outputs
C. To let agents decide freely who should speak next
D. To remove the need for validation

5. In the data-quality gate example, which field controls whether the workflow trains a model or requests data cleaning?

A. dataset_name
B. missing_rate
C. is_valid
D. n_rows

6. What is the main role of a supervisor workflow?

A. To execute all tasks in parallel
B. To route a task to the most appropriate specialist
C. To stop the workflow after a fixed number of messages
D. To replace all specialist agents with one large model

7. Which pattern is most suitable when several independent reviews can happen at the same time and then be combined?

A. Sequential workflow
B. Parallel workflow
C. Round-robin orchestration
D. Handoff orchestration

8. What is the key characteristic of round-robin orchestration?

A. Agents take turns in a fixed repeating order

B. One supervisor routes every task
C. All steps are predefined as a graph
D. Agents are not allowed to share context

9. Why are termination conditions important in autonomous multi-agent orchestration?

A. They make the model more creative
B. They prevent conversations from running indefinitely
C. They remove the need for agent instructions
D. They force all agents to produce the same answer

10. Which statement best compares workflow patterns and autonomous orchestration patterns?

A. Workflows are more flexible, while autonomous orchestration is always deterministic
B. Workflows use explicit developer-defined control, while autonomous orchestration allows more runtime agent-driven coordination
C. Both patterns always require the same number of agents
D. Autonomous orchestration does not need monitoring or debugging

Answer key: 1-B, 2-C, 3-B, 4-B, 5-C, 6-B, 7-B, 8-A, 9-B, 10-B